

Roll No.

Paper Code: TCS-408

B.Tech End Semester (Back) Examination, Dec,2022
4th Semester

Paper Name : Java Programming

Time: Three Hours

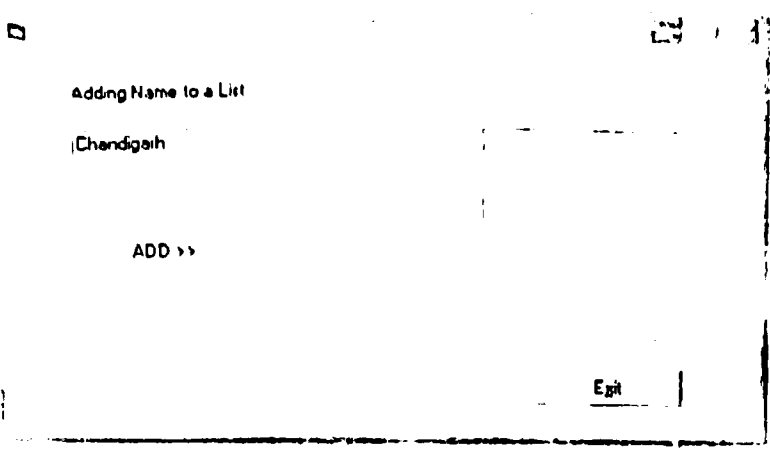
MM: 100

Note:

- (i) This question paper contains five questions.
- (ii) All questions are compulsory.
- (iii) Instructions on how to attempt a question are mentioned against it.
- (iv) Total marks assigned to each question are twenty.

Q1	(10X2=20 Marks)	
(a)	Discuss the internal architecture of JVM with help of a diagram.	CO1
(b)	List & explain the features of Java.	
(c)	Discuss the importance of string constant pool. Also, compare StringBuffer with String classes.	
Q2	(10X2=20 Marks)	
(a)	What do you mean by run time polymorphism? How " <i>one interface multiple method</i> " form of polymorphism can be achieved. Explain with suitable programming code.	CO2
(b)	Differentiate between the following: 1) Abstract Class vs Interface 2) Overloading and Overriding	
(c)	Design a class to represent bank account. Includes the following members: • Name of depositor • Account number	

	- Type of account - Balance amount in the account Methods: - To assign initial values - To deposit an amount - To withdraw an amount after checking balance. - To display the name and balance. Write a program to incorporate the constructor to provide initial values, use this keyword and also instantiate its object.	
Q3	(10X2=20 Marks)	CO3
(a)	Write a program to create two threads, one thread will print odd numbers and second thread will print even numbers between 1 to 20 numbers.	
(b)	Explain the concept of I/O streams and different streams including character stream, byte stream and file stream. Explain the concept of I/O streams and different streams including character stream, byte stream and file stream.	
(c)	Define an exception called "MyException" that is thrown when a string is not equal to "India" or "INDIA". Write a program that uses this exception.	
Q 4	(10X2=20 Marks)	CO4
(a)	Write a program to create GUI that accepts a city name from the user (through textbox) and adds it to a list box by clicking on the add button. Also write an event handler code for add>> and Exit buttons. Its screenshot is shown below:	

		
(b)	Write the differences between AWT & Swing in Java? Write suitable code to create two text fields to take two integer inputs a,b and a button (submit) in a frame. When the button gets clicked the $(a^2 - b^2)$ operation will be performed and the result will be shown in the frame.	
(c)	What is Collection and Generic Framework in Java? Write java code to insert 10, 20 at the end and 30 at index 2 of an ArrayList<Integer> having elements 1,2,3,4,5.	
Q 5	(10X2=20 Marks)	
(a)	Explain the following: i) Event Delegation Model ii) Adapter Classes	
(b)	What are JDBC drivers in Java? Write about at least two such drivers with suitable diagrams with their unique roles in database connectivity.	CO5
(c)	Describe the various steps for using JDBC? Write a program to demonstrate each step with suitable example.	